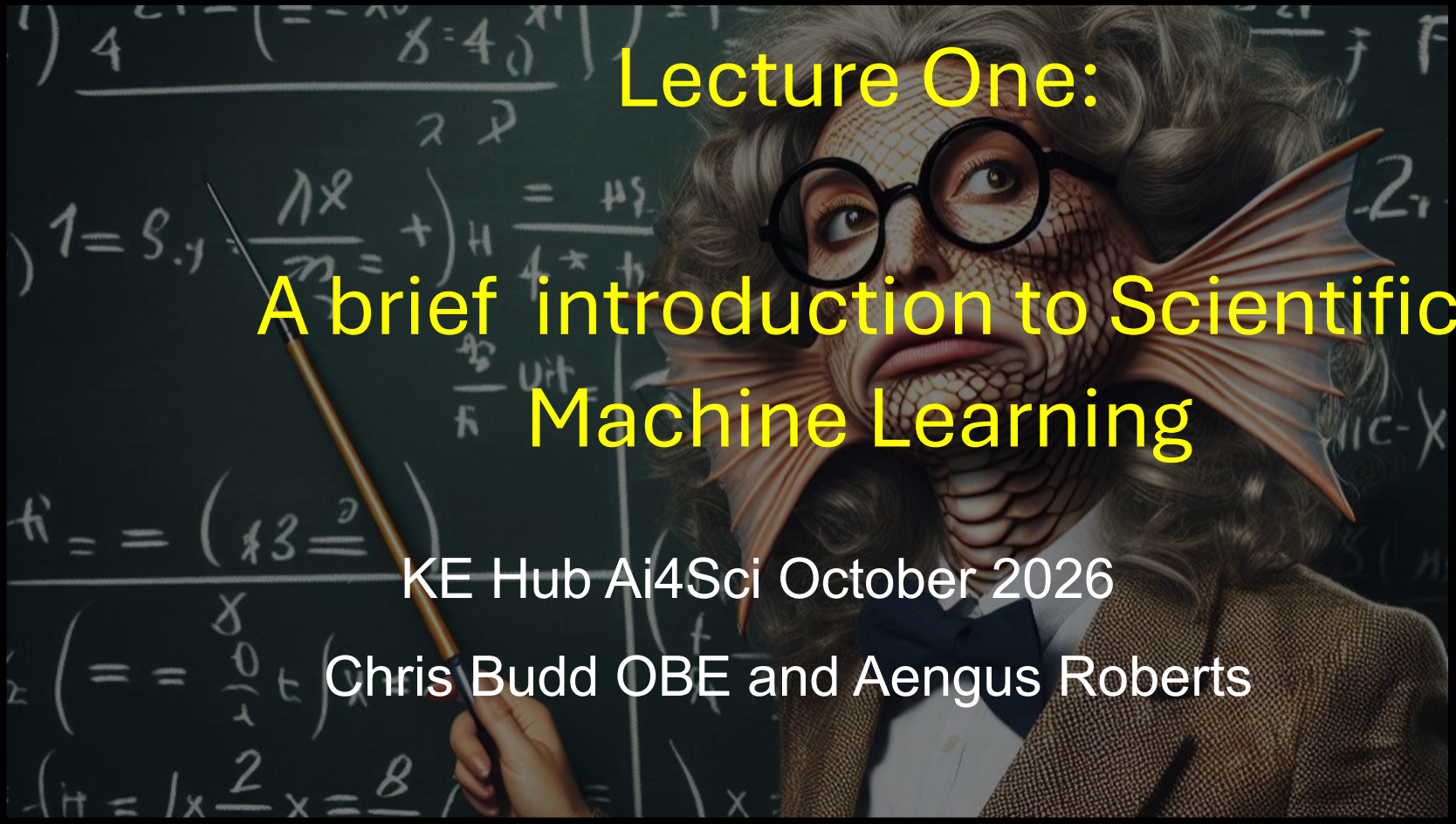


Lecture One:

A brief introduction to Scientific Machine Learning

KE Hub Ai4Sci October 2026

Chris Budd OBE and Aengus Roberts

Knowledge Exchange Hub for Mathematical Sciences

m4DL

UNIVERSITY OF BATH

www.bbc.co.uk/news/science-environment-67383755

NEWS

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Science & Environment

AI could predict hurricane landfall sooner - report

14 November 2023 · 45 Comments



AI In the News

Makes ‘better prediction of Hurricane Lee’ than any other method

How good is the hype?

Deep Learning



If you were ill, would you trust a diagnosis only made by a machine without involving a doctor?

Would you trust a weather forecast made by a machine without using the laws of physics?



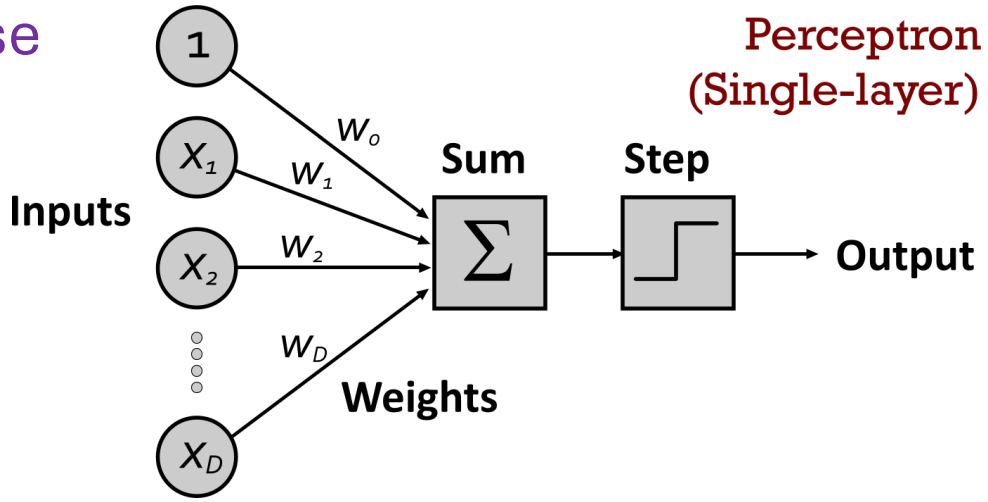


Would you be happy to fly on aeroplane designed just from looking at previous aeroplanes?

All of the above are possible now, or in the near future, thanks to machine learning. But should we trust them?

Poster by Helena Lake

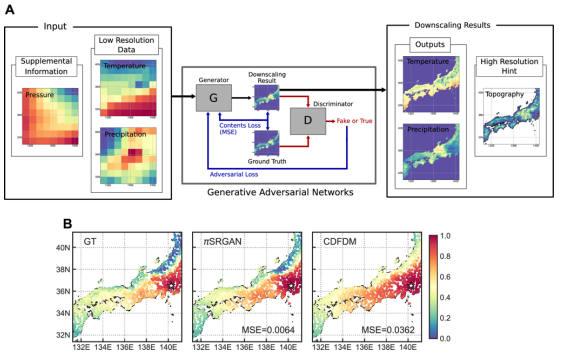
Structure of the mini-course



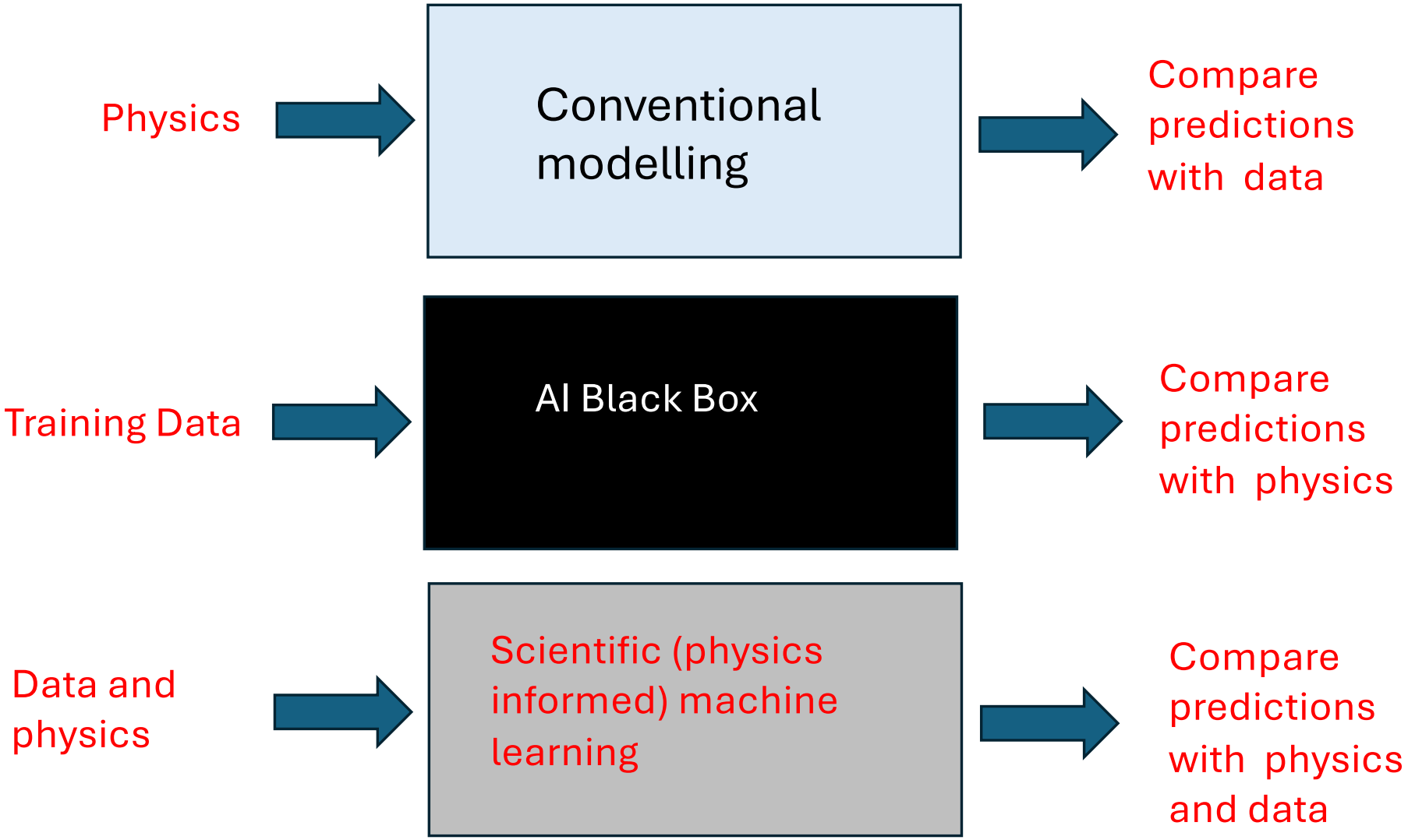
1. Introduction
2. Architectures
3. ML for differential equations
4. Neural operators and the weather
5. Neural ODES

6 Case Study: PINNS

We are seeing a major impact of machine learning mathematics, physics, engineering, and business



- Classification and identification
- Solution of differential equations (ODEs and PDEs)
- Simulation of differential operators
- Solution and regularization of inverse problems
- Prediction and time series analysis
- Classification and generation of images (down scaling)
- CAD
- Forecasting the weather (from data alone!)
- Proving theorems!

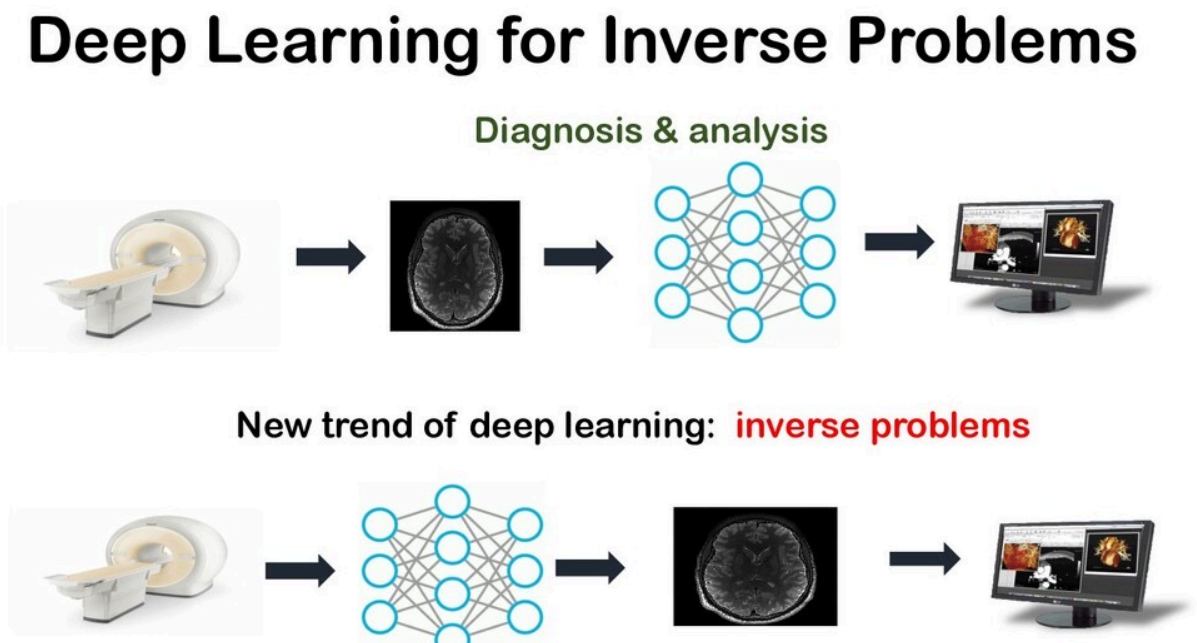

Physics
Conventional modelling
Compare predictions with data
Training Data
AI Black Box
Compare predictions with physics
Data and physics
Scientific (physics informed) machine learning
Compare predictions with physics and data

Scientific Machine Learning

Aims:

- Derive

- Effective
- Efficient
- Transparent
- Reliable
- Unbiased (data, inductive, learning)



Machine learning methods informed by physics and other disciplines

- Prove (where possible) rigorous results on their accuracy, stability, and convergence

- Apply them to reliably solve important scientific and engineering problems

A brief introduction to the world of (scientific) machine learning

Image classification/Prediction

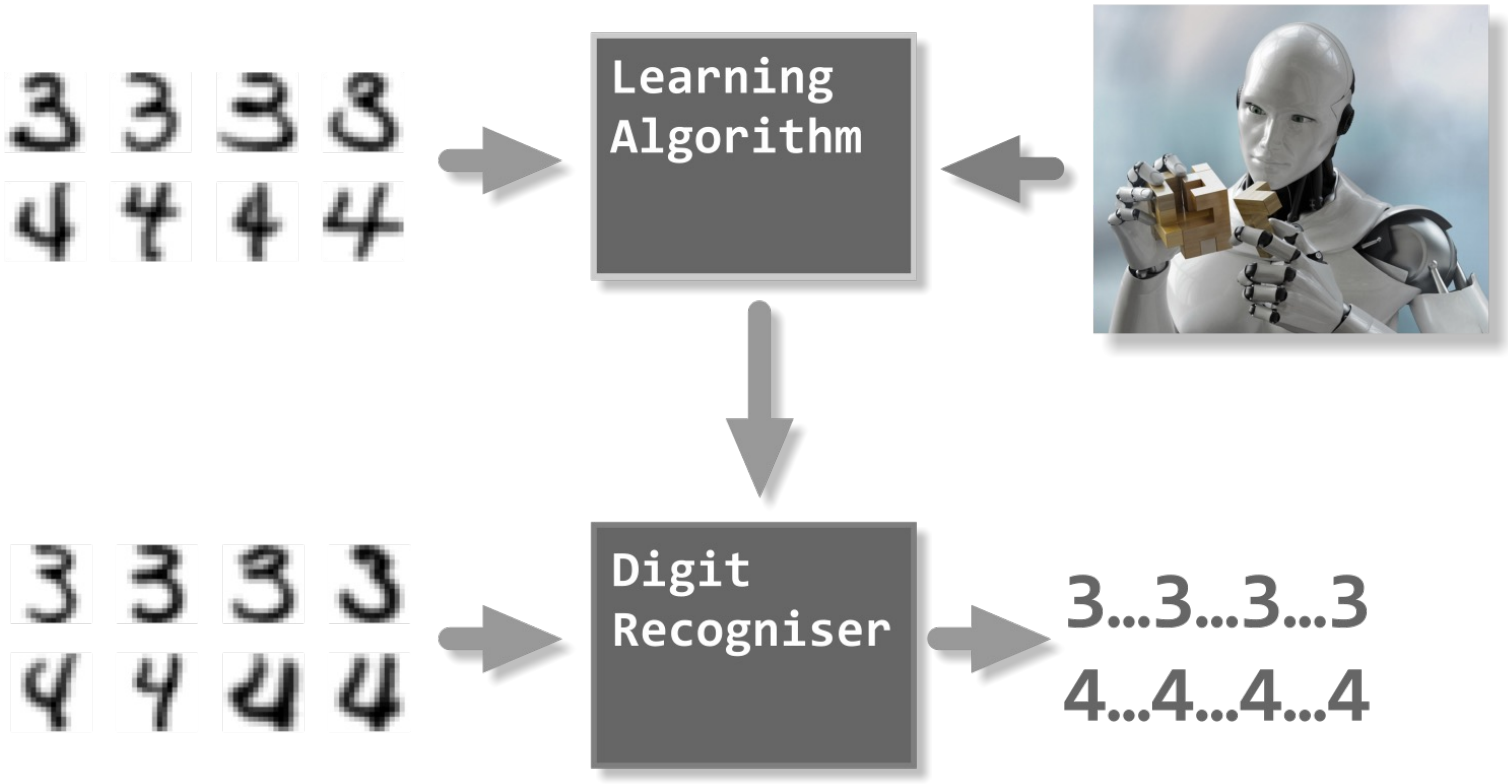
Probability distribution of images



Neural net trained on samples from the set of images

→ Image labels

Operation of a classifier



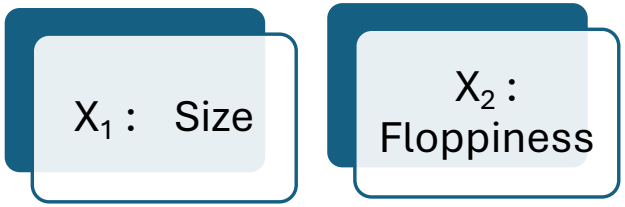


Example: Can you tell a cat from a dog?

Feature identification

One measurement is not good enough

Make two measurements

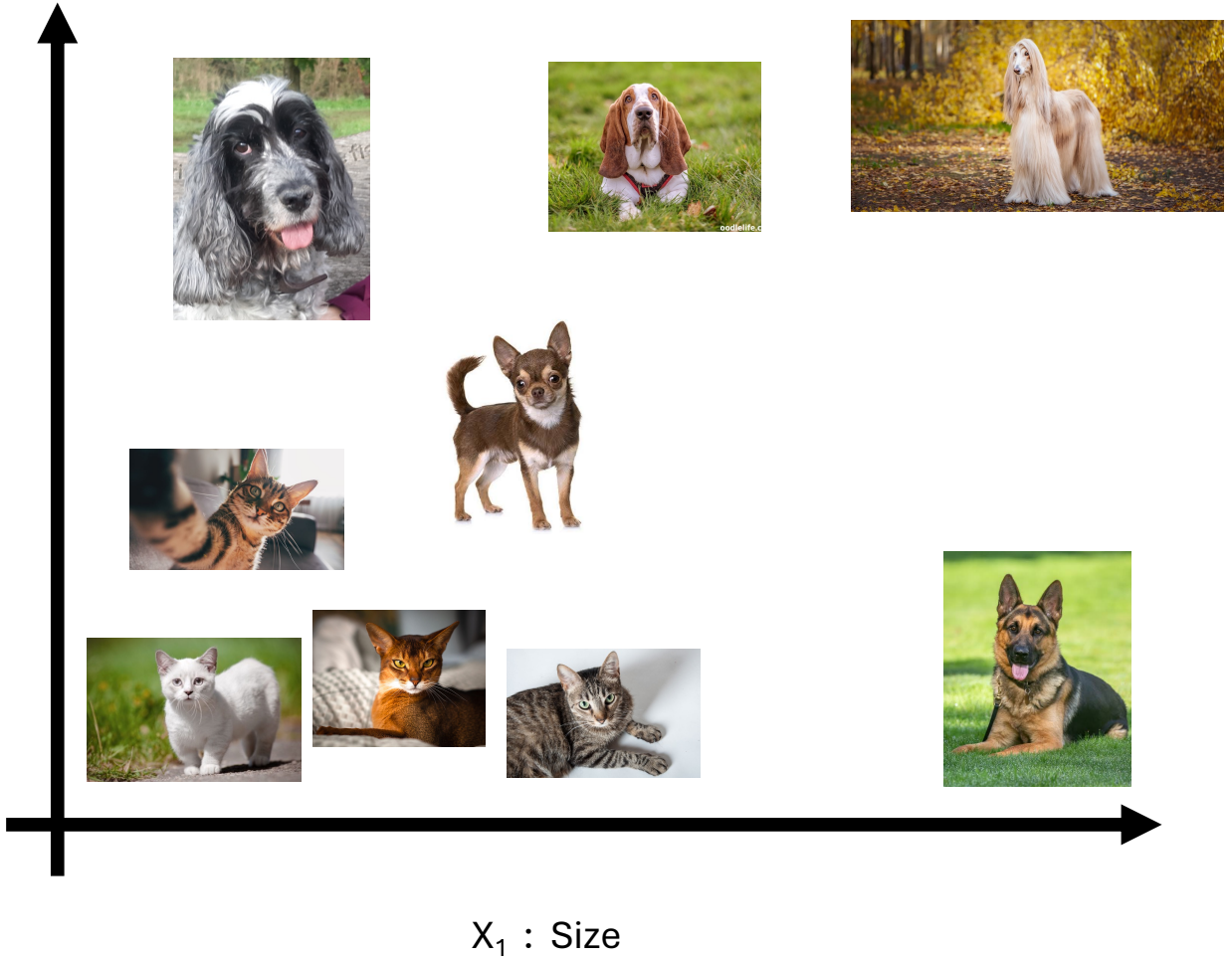




Measure size and floppiness for lots of cats and dogs and plot

X_2 :Floppiness

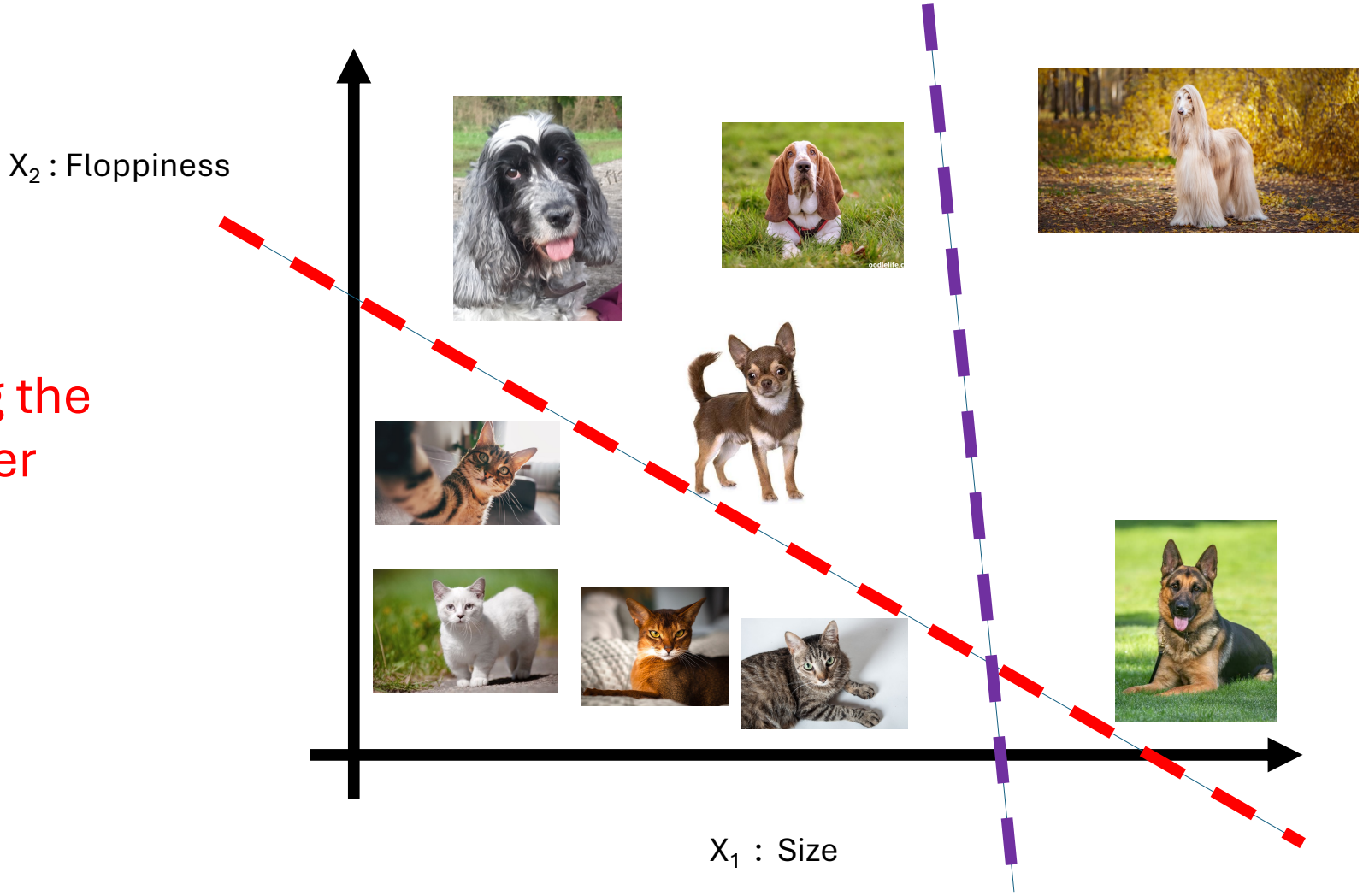
Training Data



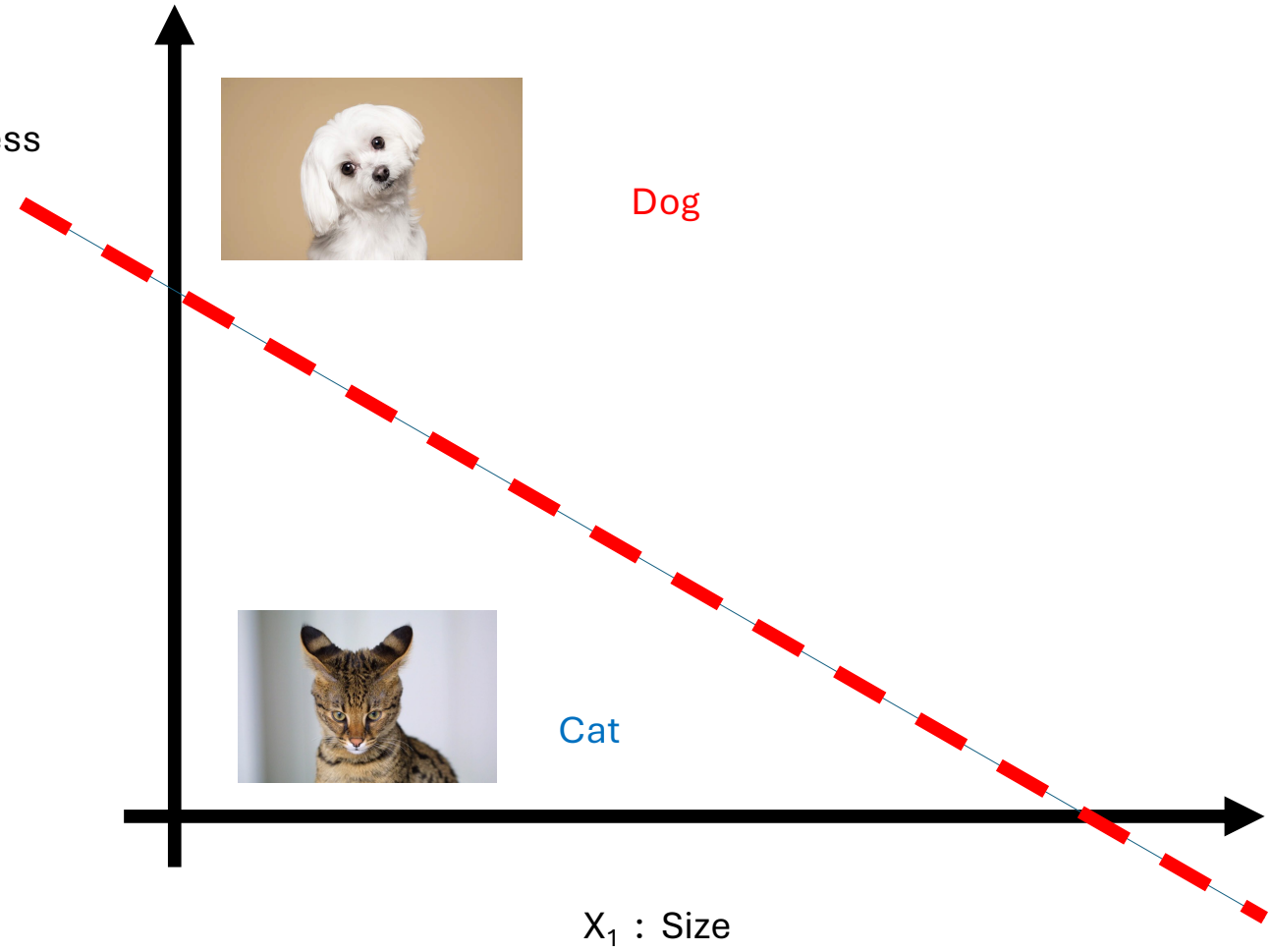
X_1 : Size

Draw a line which separates cats from dogs

Training the classifier



Can now tell a cat from a dog



Using the classifier

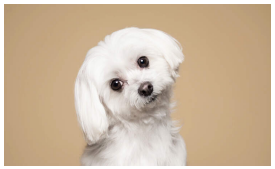
What is going on?

Equation for the line is:

$$w_1X_1 + w_2X_2 - C = 0$$

$$w_1X_1 + w_2X_2 - C > 0 \ : \ \text{Dog}$$

$$w_1X_1 + w_2X_2 - C < 0. : \ \text{Cat}$$



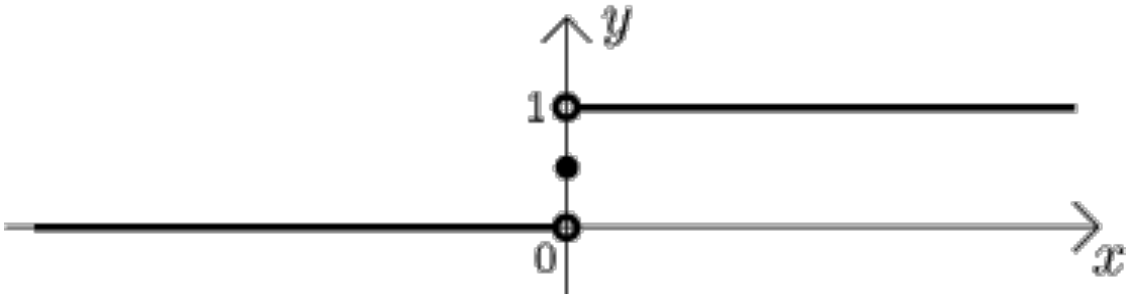


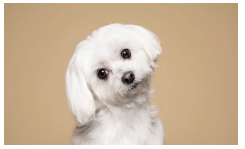
Train the weights w_1, w_2, C on the data to give the right result

$$Out = H\left(\sum_{i=1}^{D} w_i\, X_i - C\right)$$

$$H(x) = \begin{cases} 1, & x > 0; \\ \frac{1}{2}, & x = 0; \\ 0, & x < 0. \end{cases}$$



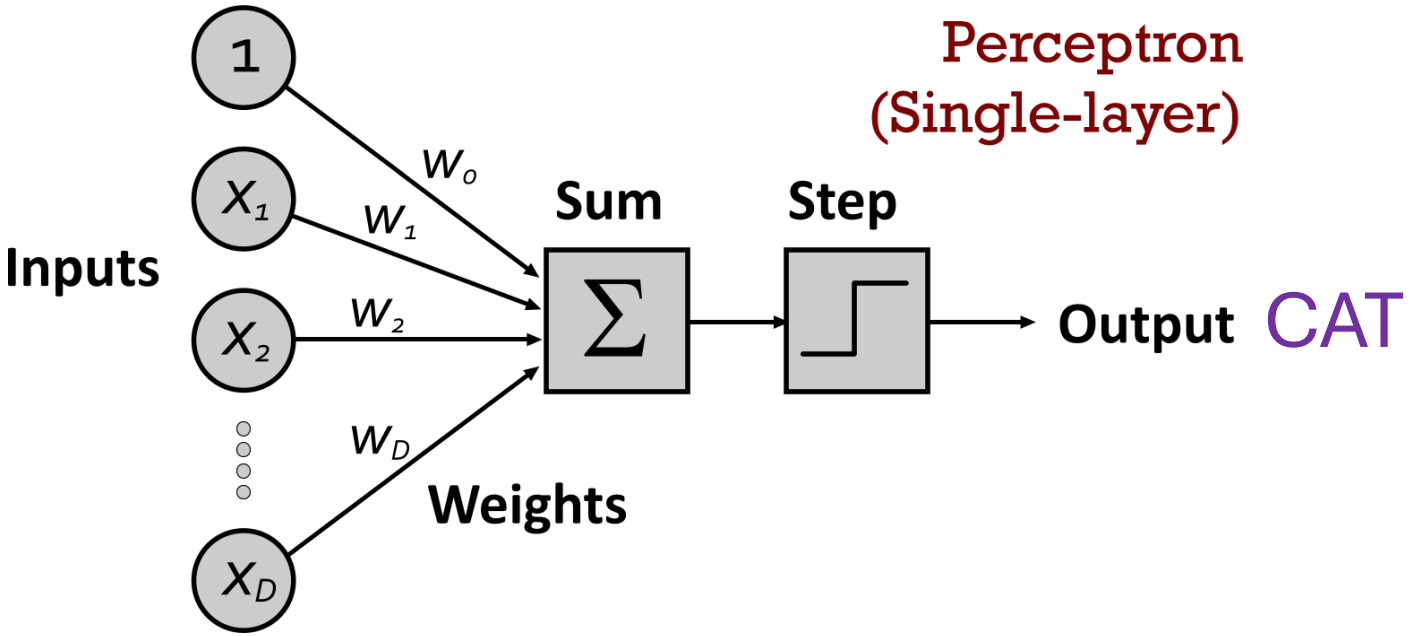




Extract important features

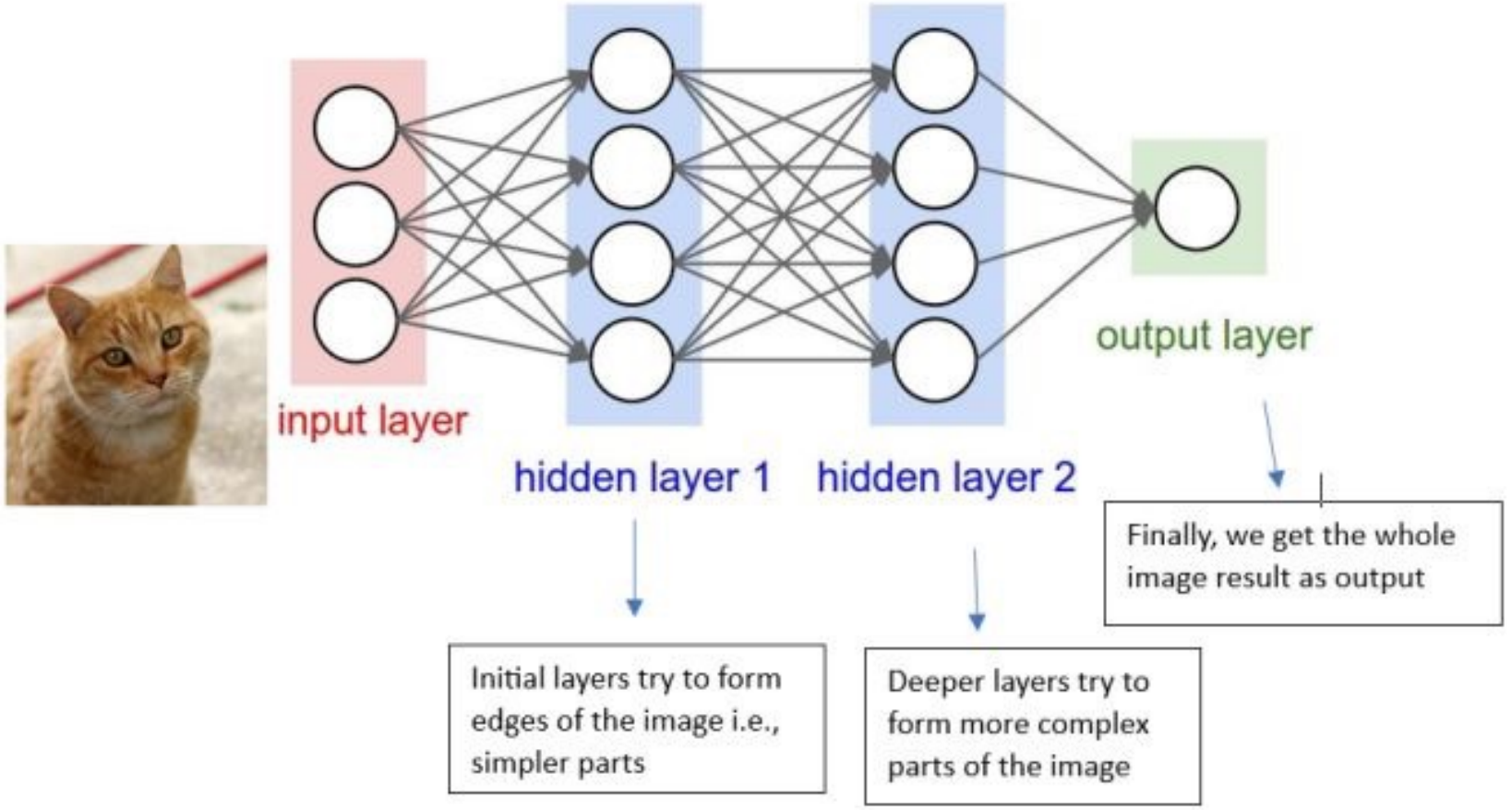


Image

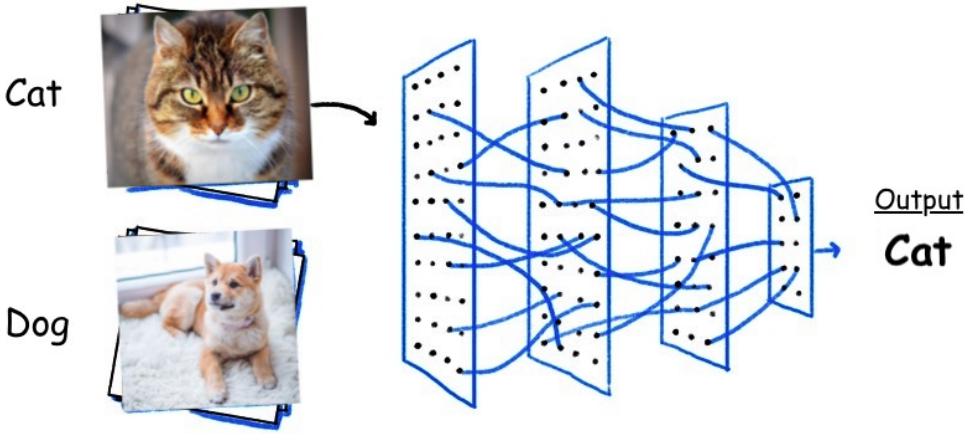


Basic architecture of a neural net classifier (1963)

Deep neural nets (2010) use many more layers and can be used to make much more complex decisions






Cat
Dog
Output
Cat



Or more mathematically, using a notation we use later on:

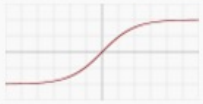
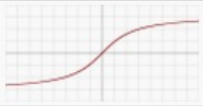
$$y_{j+1} = \sum_{i=0}^{W-1} c_i \; \sigma_{i,j} \left(a_{i,j} \; y_j + b_{i,j} \right)$$

Input: $y_0 = x$. Output: $y(x) = y_L$

W: Width L: Depth

$\sigma_{i,j}$: Activation functions. (usually nonlinear)

θ: Set of All of the parameters

Name	Plot	Equation	Derivative
Identity		$f(x) = x$	$f'(x) = 1$
Binary step		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ ? & \text{for } x = 0 \end{cases}$
Logistic (a.k.a Soft step)		$f(x) = \frac{1}{1 + e^{-x}}$	$f'(x) = f(x)(1 - f(x))$
TanH		$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1$	$f'(x) = 1 - f(x)^2$
ArcTan		$f(x) = \tan^{-1}(x)$	$f'(x) = \frac{1}{x^2 + 1}$
Rectified Linear Unit (ReLU)		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
Parameteric Rectified Linear Unit (PReLU)[2]		$f(x) = \begin{cases} \alpha x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} \alpha & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
Exponential Linear Unit (ELU)[3]		$f(x) = \begin{cases} \alpha(e^x - 1) & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} f(x) + \alpha & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
SoftPlus		$f(x) = \log_e(1 + e^x)$	$f'(x) = \frac{1}{1 + e^{-x}}$

Some Definitions

Architecture: The overall structure, and the way that the layers are connected together

Expressivity: How well a perfectly trained network is able to do the task it is designed for

Trainability: Training a neural network is the process of finding the parameters θ so that it does a pre assigned task. Trainability is a measure of how well we are able to do this.

*Transparency: H*ow well we understand what the network is doing, and the impact of the different parameters in the network on the output

Training a network

Need to find the best values for the Parameters

Achieve this by training the NN using a training set of data and a measure of success

Can take many hours of computer time on a GPU.

With a vast expense of energy!!



- Supervised training

Have a labled training set

eg. cats + dogs, musical genres, Inverse problems, approximation

- Reinforcement training

Have a measure of success

eg. winning GO, solving an equation (eg. a PINN), variational method



Mathematical Ingredients of the training process

Loss function L(W) : How close is the answer

Back propagation: How does the answer depend on changes in the parameters.

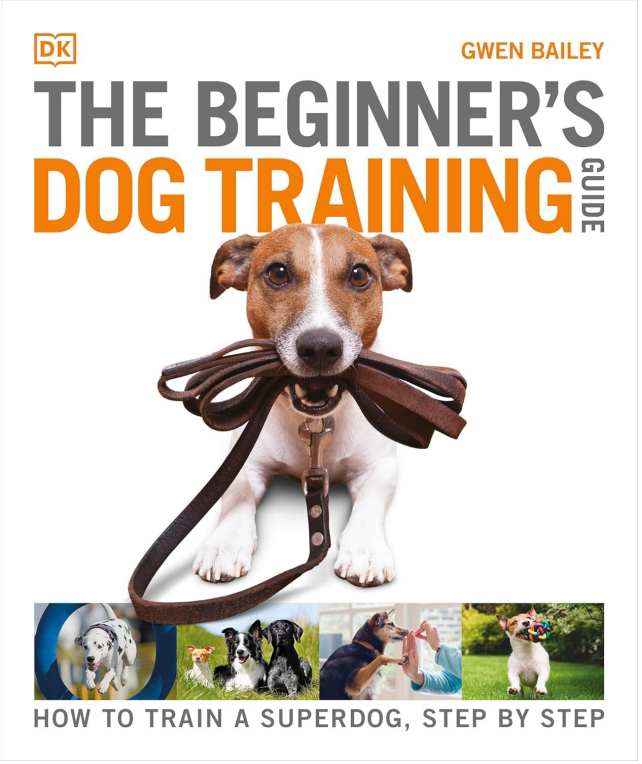
Achieved by using automatic differentiation

PyTorch does this with ease!

Optimisation: Choose the parameters to

minimize the loss over (a subset of the) training data

Validation: Test on separate data to avoid over fitting to the answer



Optimise the weights using optimisation methods:

Eg. stochastic gradient descent, ADAM, quasi-Newton, simulated annealing, Monte-Carlo Tree Search (MCTS)

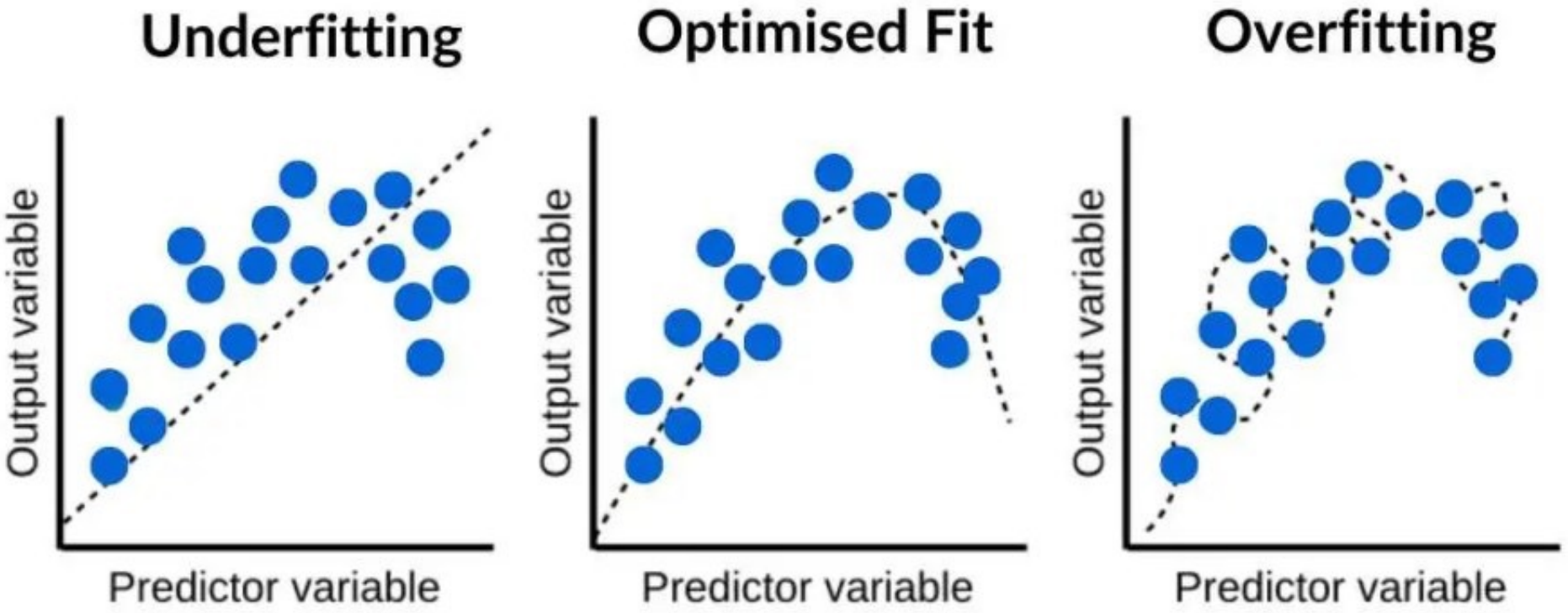
Implemented in eg. PyTorch

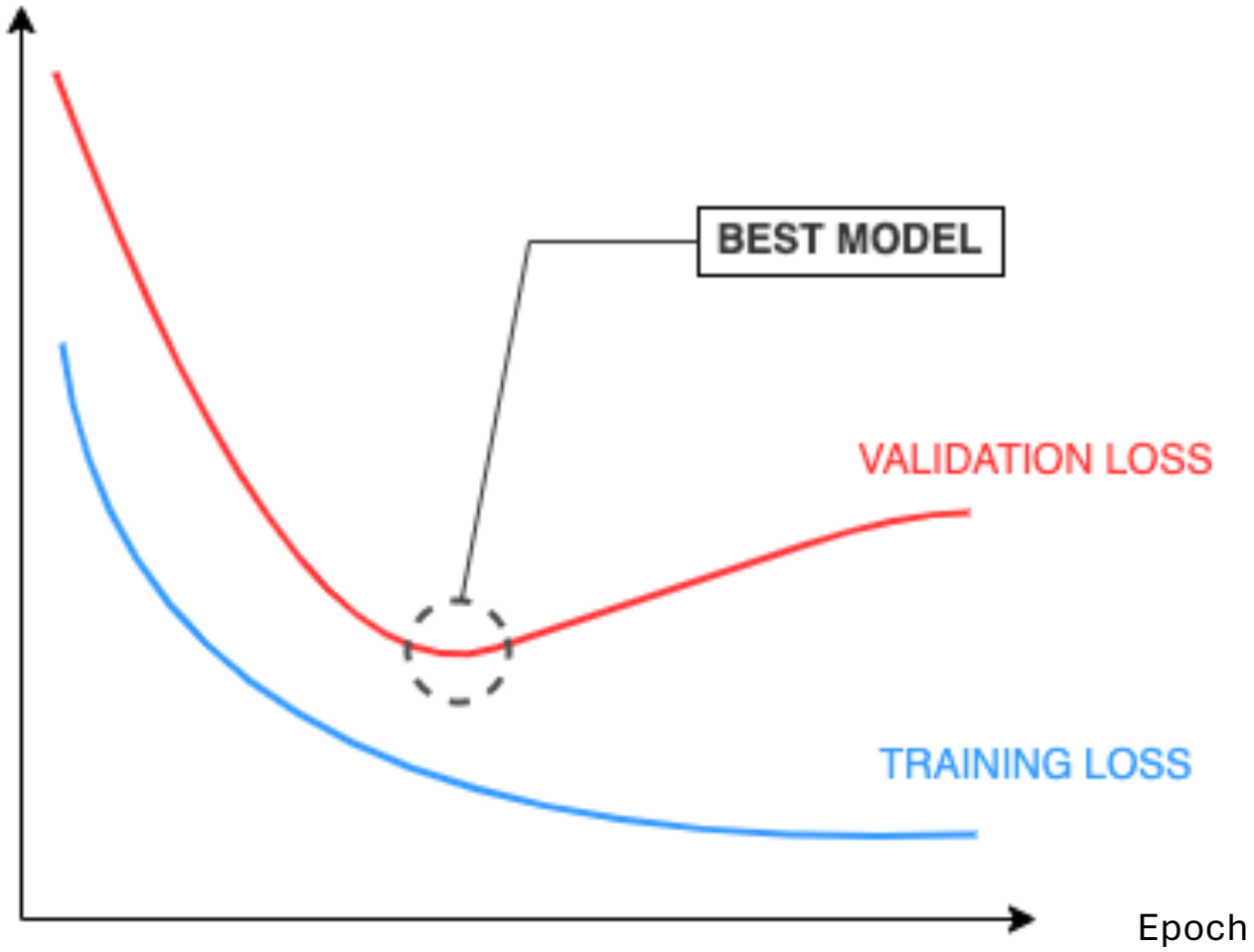
(see the Case studies)

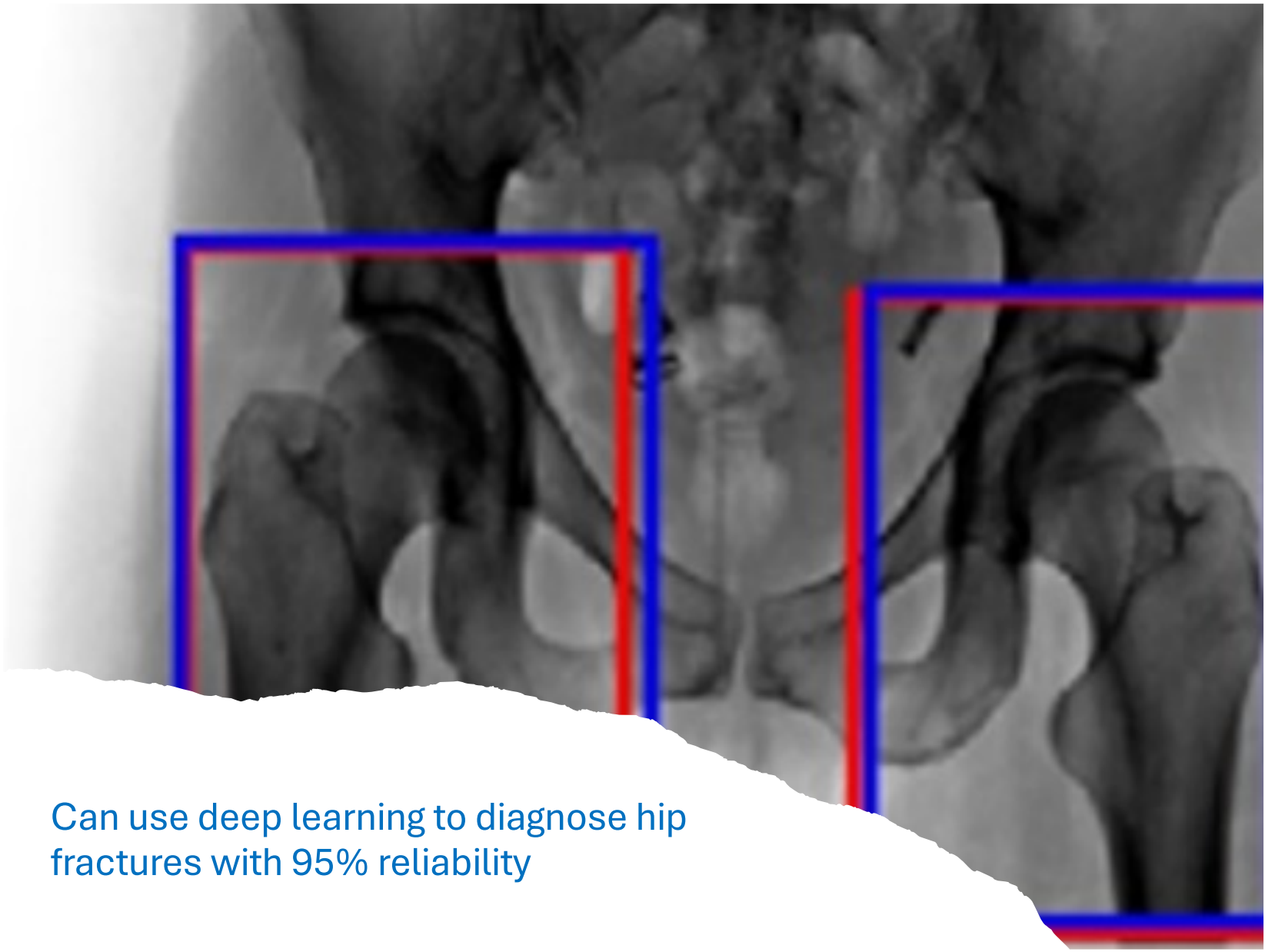


Tensor Flow/Keras

AlphaZero used reinforcement learning over 700,000 games with MCTS optimisation

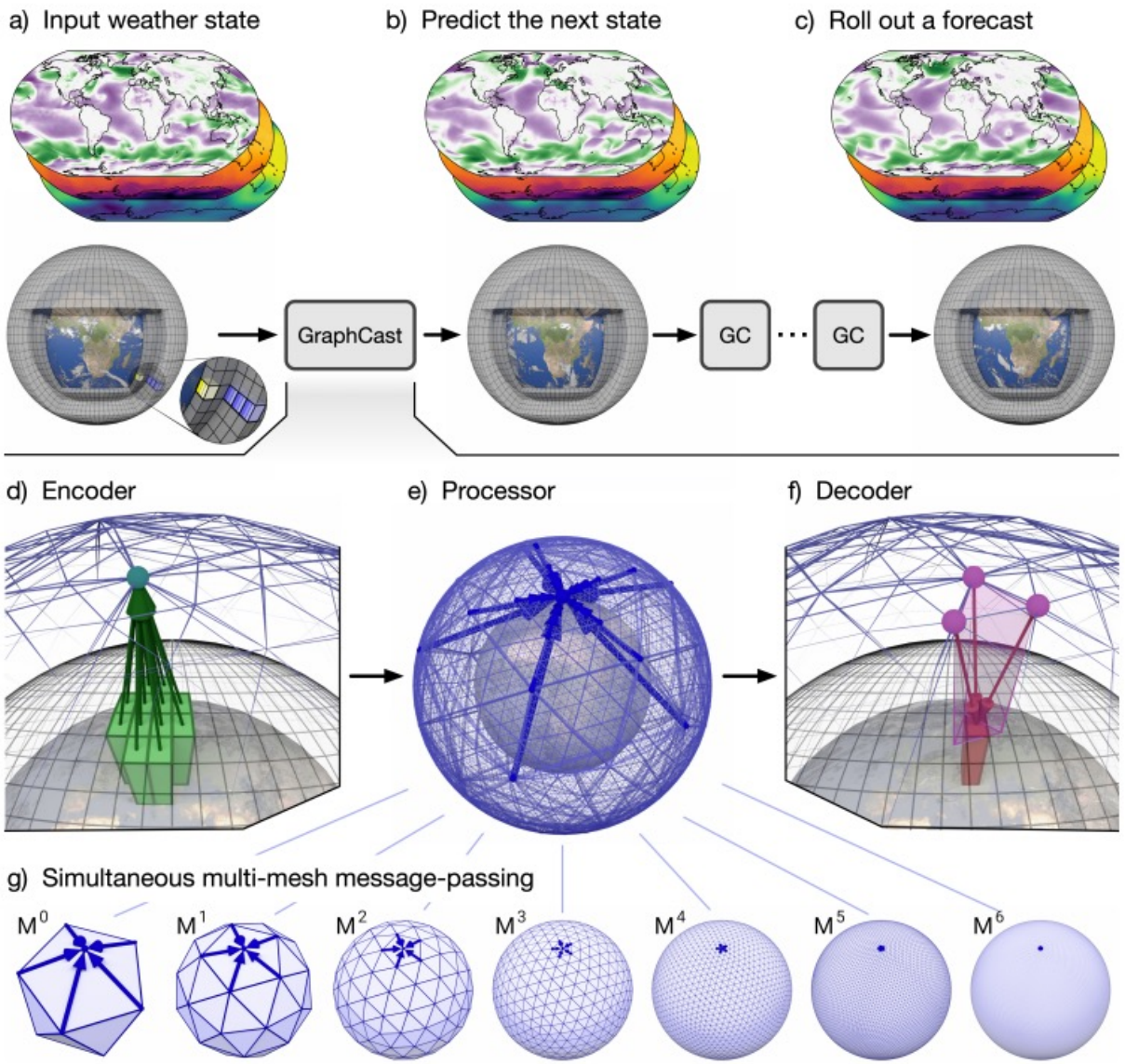

Underfitting
Output variable
Predictor variable
Optimised Fit
Output variable
Predictor variable
Overfitting
Output variable
Predictor variable


BEST MODEL
VALIDATION LOSS
TRAINING LOSS
Epoch



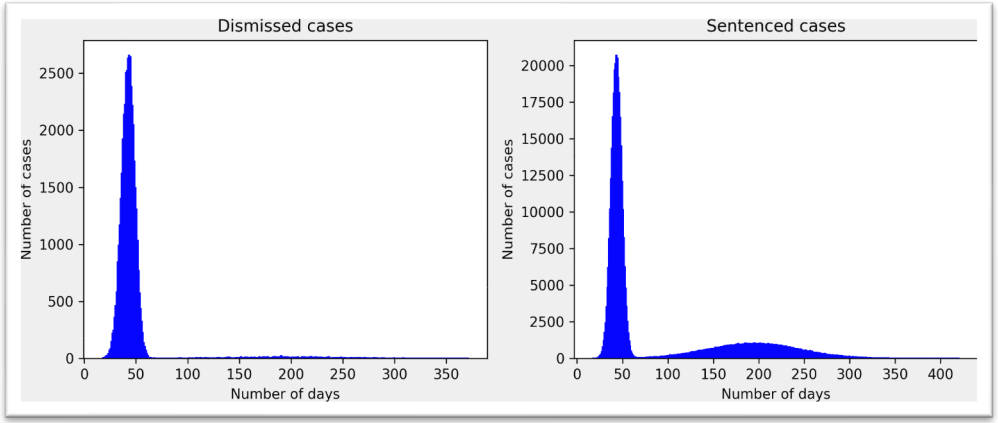
Can use deep learning to diagnose hip fractures with 95% reliability

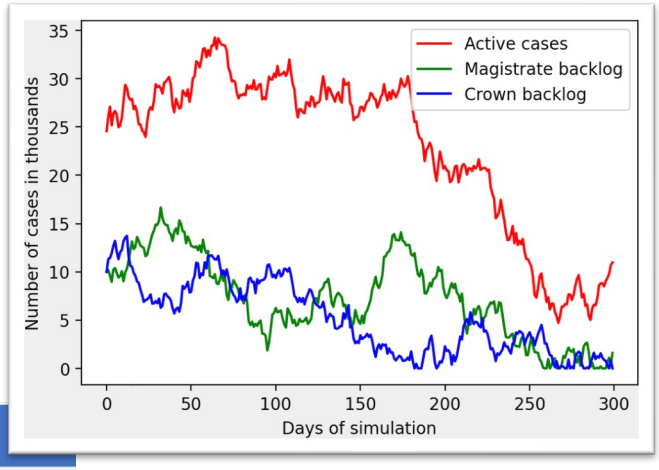
Weather forecasting trained on the ERA5 Data Set



Maths and AI for Justice

NN can predict case volume and duration in the judicial system based on case characteristics





Time distribution: Lognormal
Time spent in summoned: 34 days
Summoned to Exit: 9 days
Time spent in Sentencing: 7 days

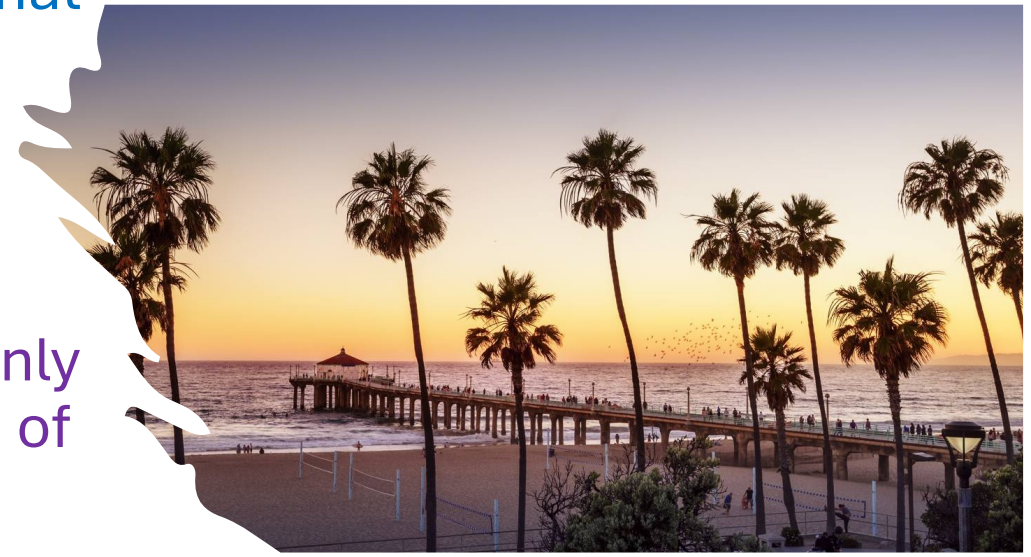
Data: the national statistics for the Criminal court statistics quarterly (April - June 2023)

But ..Can a weather forecast trained on Californian data predict a storm in the UK?



Learning only from data makes it hard to make decisions away from that data

Would you trust a computer to make a judicial judgement if only trained on some types of cases?





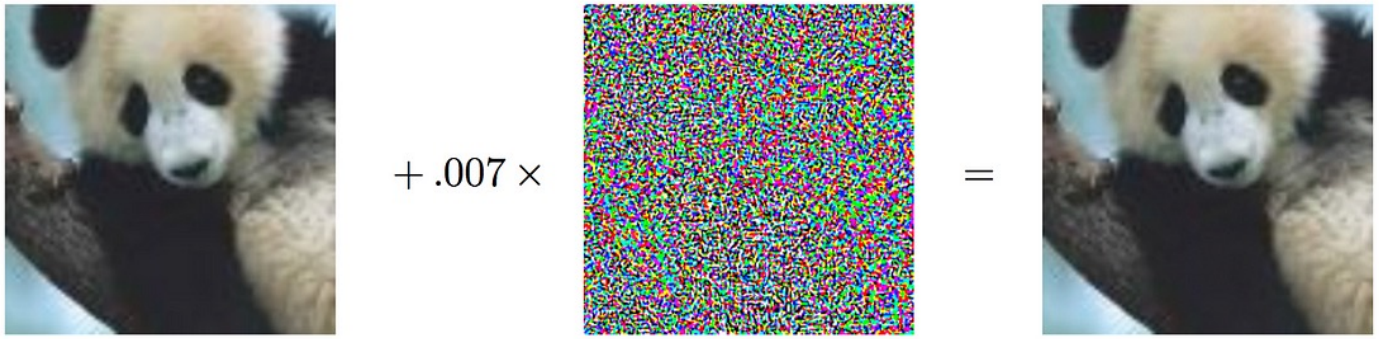
Also:

Some images are hard to classify correctly



Machines can get it wrong

Identified as a 45mph speed sign



And can get confused by an ‘adversarial attack’

Study this as a conditioning problem in numerical analysis

Machine learning models can memorise data: our privacy is at risk

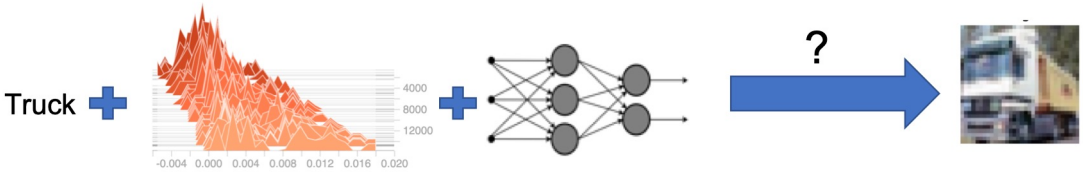
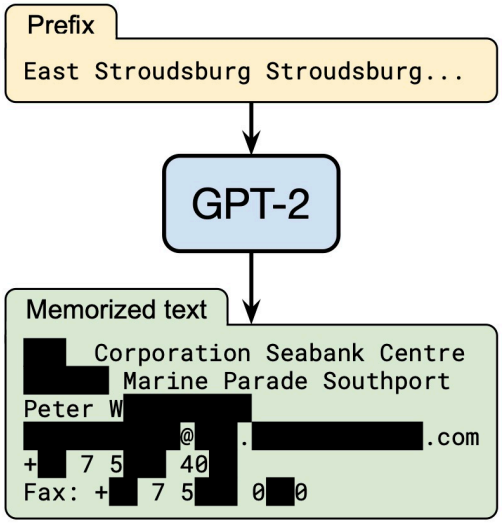


Image classification models can memorise their training images through the training updates (i.e. gradients).

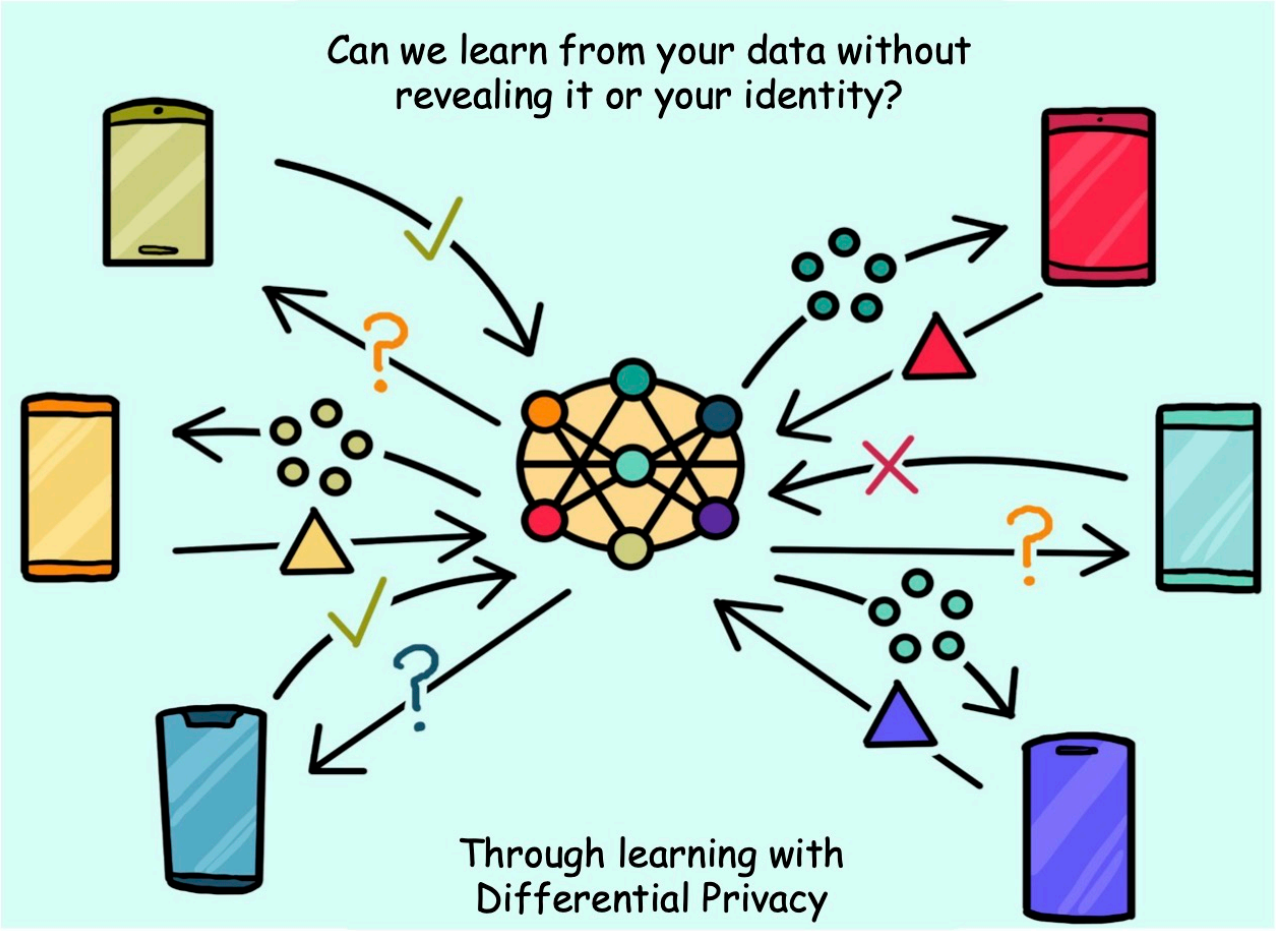


Language models can also memorise their training can be extracted using text prompts.

Chen C, Campbell N. Analysing Training-Data Leakage from Gradients through Linear Systems and Gradient Matching, The 33rd British Machine Vision Conference (BMVC), 2022

Carlini, Nicholas, et al. "Extracting training data from large language models." 30th USENIX Security Symposium (USENIX Security 21). 2021.

One way to mitigate the privacy risk: training with differential privacy



Maths is our best way of explaining how AI is working and answering questions about its limitations and its possibilities

Computational Mathematics

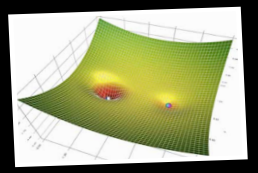
Work by 19th century mathematicians **Charles Babbage** and **Ada Lovelace**, and 20th century mathematicians **Alan Turing** and **John Von Neumann** leads to the invention of the modern computer







20th century developments in the **mathematics of optimisation** led to the rapid growth of machine learning



Statistics and Data Science

19th century mathematicians **Thomas Bayes** and **Carl Friedrich Gauss** transformed the way that we understand and manipulate data





$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \left[\sum_{i=1}^{N} (f_\theta(x_i) - y_i)^2 + R(\theta) \right]$$

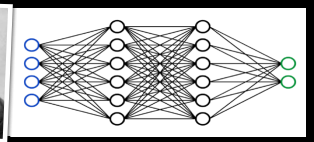
This mathematics is used in medical imaging



Algebra and Calculus

Matrices, invented by 19th century mathematician **Arthur Cayley**, provide the structure for the Internet and all AI architectures





$$\boldsymbol{x}_{k+1} = \sigma\left(A_k \boldsymbol{x}_k + \boldsymbol{b}_k\right) \in \mathbb{R}^W$$



20th century mathematician **Kiyoshi Itô** developed stochastic calculus, the basis of generative AI

